

Auditing Search-Budget Portability Across Graph-ANNS Rebuilds: [Experiments & Analysis]

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Abstract

Graph indexes are rebuilt to maintain search quality, but the budget policy that serves queries is often retained as reusable state. We study when that policy remains portable, how target evidence should qualify reuse, and when recovery repays its information cost. Across unchanged SIFT-100K and Arxiv-Nomic-100K collections, transferring retrospective first-passing budgets between hnsplib and Faiss builds adds 17.17–23.60 percentage points of query failure relative to target-specific references. After a 5% data refresh, an unchanged recurring-query policy crosses the 5% risk limit on both datasets. Index-Conditioned Budget Audit (ICBA) turns these measurements into a decision contract that separates candidate construction, target certification, fallback, held-out evaluation, and acquisition cost. Within this contract, target-calibrated pooling (TCP) retains jointly controlled mean-distance reductions of 44.39% and 29.86%; cold-history acquisition amortizes near 396,000 and 457,000 executions. A second route fixes a source-derived one-rung candidate before target access. Fresh target roles qualify all 1,104 Faiss source–target decisions, with 0.13%/0.53% risk and 8.28%/29.08% mean-distance savings. A preregistered hnsplib Deep1M replication likewise qualifies all 56 decisions and saves 34.28%, without increasing pooled p95. These results establish budget portability as an index-conditioned property and show that target qualification can recover search work across implementations and scale, while complete lifecycle value remains workload dependent.

Keywords

approximate nearest-neighbor search, graph indexes, index rebuilding, risk auditing, budget transfer

1 Introduction

A graph index and its search-budget policy jointly determine the quality and cost of approximate nearest-neighbor search (ANNS). Hierarchical Navigable Small World (HNSW) graphs and Vamana make this tradeoff central to efficient vector retrieval [12, 14]. Index maintenance changes the execution environment of that policy. Construction order and randomization can alter traversal even when vectors and queries are unchanged; data refresh can also change nearest-neighbor truth. A budget retained across rebuilds is therefore a portability decision, not inert configuration.

Portability matters at the query-risk level. We measure the probability of Recall @10 < 0.95: because recall is discrete, one missing neighbor is a failure. A 5% risk limit asks that at least 95% of queries recover all ten neighbors. This requirement differs from mean recall of 0.95 and from the expected missed-neighbor fraction. It also makes target difficulty visible: a transferred action can fail because the target itself contains unresolved queries or because the source action adds loss beyond the target-specific reference.

Terms at a glance. Tables and figures shorten SIFT-100K and Arxiv-Nomic-100K to SIFT and Arxiv.

Term	Meaning
ICBA	Index-Conditioned Budget Audit: the measurement and decision workflow.
TCP	Target-calibrated pooling: recurring-query recovery from pooled history.
NDC	Number of distance calculations; not elapsed time.
CP / UCB	Clopper–Pearson / upper confidence bound.
Transport risk	Target failure under a transferred action or policy and a declared event.
LOTO	Leave one target build out.
Unresolved label	No registered action reaches the recall target.
Endpoint	Largest grid action, whose risk is measured.

We build the evidence in three stages. First, on unchanged SIFT-100K and Arxiv-Nomic-100K collections, we transfer each query’s retrospective first-passing action across hnsplib and Faiss rebuilds. Additional target failure ranges from 17.17 to 23.60 percentage points across all four implementation–dataset blocks. Second, after a 5% data refresh, we replay an executable recurring-query policy without changing its source information. Its risk rises from 3.77% to 6.69% on SIFT and from 3.19% to 7.24% on Arxiv-Nomic. Third, we freeze source-derived recovery candidates and reserve fresh queries for target certification and evaluation. This stage qualifies all 1,104 Faiss decisions across the two 100K datasets and all 56 hnsplib decisions in a preregistered Deep1M replication, while retaining positive mean and tail results.

Index-Conditioned Budget Audit (ICBA) organizes these observations as a deployment decision. It fixes the permitted observations and failure event, separates candidate construction from target certification and evaluation, includes fallback in the completed outcome, and prices information acquisition. The resulting contract prevents response evidence, statistical qualification, serving-work savings, and lifecycle value from being treated as the same quantity.

We instantiate two recovery routes within that contract. *Target-calibrated pooling* (TCP) pools historical budgets for recurring profiled queries and selects one target shift before independent certification. Under a joint 5% confidence-error allocation, it retains mean-distance reductions of 44.39% on SIFT and 29.86% on Arxiv-Nomic. The fixed-slack route removes target selection: it freezes a one-rung source-derived candidate, then uses fresh target queries only to certify and evaluate that candidate. Its source-only stage exposes an implementation boundary on the 100K grids; its target-certified Faiss and Deep1M replications establish recovery across implementations and scale. Figure 1 summarizes the two portability tests that motivate these decisions.

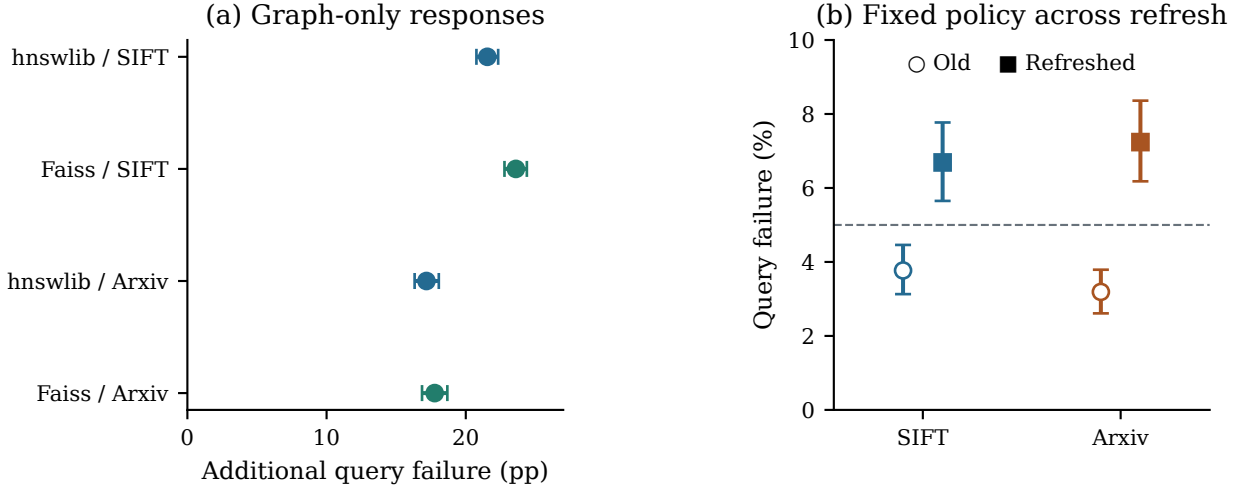


Figure 1: Two portability tests. Left: graph-only transfer of retrospective first-passing actions; additional failure above each target’s own first-passing reference, with 95% query-cluster intervals conditional on 24 builds. Right: the same executable history-pool policy on matched old and 5%-refreshed builds; average query risk over ten builds, with crossed target-build/query 95% intervals. The old/new comparison changes both graph and data. Directed build pairs remain grouped within their shared query and target units.

We make three contributions. First, a layered empirical study isolates graph-only response change, matched refresh-policy deterioration, and target-certified execution across hnswlib, Faiss, two data families, and 100K-to-1M scale. Second, ICBA supplies a reusable contract for observations, query roles, risk qualification, fallback, and cost; its fixed-target certificate states exactly what a target decision authorizes. Third, TCP and the fixed-slack route demonstrate complementary recovery regimes. TCP serves recurring profiles with target selection, while fixed slack uses a frozen source candidate followed by target certification. Their positive and endpoint-clipped cases identify where each route retains useful work and when its information cost amortizes. DARTH, Ada-ef, and Vamana analyses extend the scope without forming a common leaderboard.

The systems implication is concrete: version a search-budget policy with the index, workload, and calibration record. Reuse requires target evidence, and the appropriate recovery route depends on which observations are available and how long the resulting policy will serve.

2 Related Work and Positioning

Graph construction, evaluation, and maintenance. HNSW combines hierarchical navigation with graph search, while DiskANN introduces Vamana as part of a system for large-scale search [12, 14]. Graph-search evaluations compare construction paradigms across datasets and scales [4], and insertion order and intrinsic dimensionality affect HNSW recall [8]. These studies establish construction sensitivity. We study the decision that survives construction: an action or policy transferred from a source build to a target, with target difficulty separated from additional transfer loss. FreshDiskANN supports streaming updates [13]; IP-DiskANN

performs insertions and deletions in place [16]; Wolverine repairs monotonic search paths disrupted by deletion [11]. These systems change the maintained graph. ICBA evaluates the validity of a budget decision on the graph that maintenance produces.

Adaptive work allocation and declarative recall. LAET predicts query-specific termination conditions using static and intermediate-search features [10]. DARTH predicts achieved recall during execution and adapts prediction intervals to terminate HNSW or IVF searches at a declarative target [5]. ANNiE estimates the work needed for a recall target from graph-derived features; ANNiE-S uses those estimates directly [15]. Ada-ef models query-to-dataset similarity distributions to choose an HNSW exploration parameter and updates its statistics after insertions or deletions [18]. Portability concerns the validity and cost of carrying any such decision between builds. Our frozen DARTH and Ada-ef integrations retain their permitted signals and update procedures, so they test qualification in those settings rather than define a common ranking.

Quality guarantees and their target quantities. ConANN applies conformal risk control to IVF search, controlling expected false-negative rate while adapting the number of probed clusters [9]. Its objective is the expected fraction of missed neighbors, whereas our principal event is whether a query falls below a recall threshold. ANNiE’s optional recall-target guarantee addresses the fraction of queries reaching the requested recall using quantile-loss minimization [15]. The DARTH+ preprint also introduces optional probabilistic quality guarantees for HNSW and IVF [6]. We study which build and policy an assessment covers, what information is

reused or reacquired, and how the assessed decision behaves after rebuilding. Each comparison therefore retains its original risk quantity.

Calibration, selection, and the role of ICBA. Conformal risk control bounds an expected monotone loss under its calibration assumptions [3]. Learn then Test formulates risk-controlled parameter selection as multiple hypothesis testing [2]. These are statistical foundations for auditing a search decision. ConANN explicitly aligns calibration with index construction and assumes exchangeable query data [9]; rebuilding can change the loss induced by search even when the query population is retained. ICBA makes the target build, failure event, candidate construction, certification, and fallback rule explicit. TCP and fixed slack instantiate two information patterns within that audit. This separation identifies when target information restores search efficiency and whether the resulting work saving repays its acquisition cost.

3 Portability, Risk, and the Scope of Guarantees

3.1 Execution Risk and Retrospective Responses

Let G specify a vector snapshot, graph, and search implementation, and let $\mathcal{A} = (a_1 < \dots < a_J)$ be its native action grid. A policy π maps permitted query and history observations to an action. For $q \sim \mathcal{Q}$,

$$\begin{aligned} Z_G(q, a) &= \mathbf{1}\{\text{Recall}@k(G, q, a) < \tau\}, \\ \mathcal{R}_G(\pi) &= \mathbb{E}_q Z_G(q, \pi(q)). \end{aligned} \quad (1)$$

Truth is computed against the corresponding snapshot. We distinguish target transport risk $\mathcal{R}_{G_t}(\pi_s)$, source risk $\mathcal{R}_{G_s}(\pi_s)$, and a reference-relative increment

$$\Delta_{s \rightarrow t} = \mathcal{R}_{G_t}(\pi_s) - \mathcal{R}_{G_t}(\pi_t^{\text{ref}}). \quad (2)$$

A matched old-to-new comparison instead subtracts $\mathcal{R}_{G_s}(\pi_s)$, keeping the policy fixed. These two differences answer different questions.

A retrospective first-passing label is $B_G^{\text{first}}(q) = \min\{a_j : Z_G(q, a_j) = 0\}$. A stable-tail label is

$$B_G^{\text{tail}}(q) = \min\{a_j : Z_G(q, a_\ell) = 0 \text{ for every } \ell \geq j\}. \quad (3)$$

Both require query truth; an empty set is $+\infty$, an unresolved label. Native action order need not make either raw recall or realized NDC monotone. Consequently, an action larger than a first-passing label need not succeed, whereas a finite stable-tail label establishes success above it on the recorded grid. The graph-only study uses first-passing labels and evaluates the transferred action's *actual* target response, rather than replacing execution failure by a budget-order comparison.

3.2 Information Limits Explain What Must Be Observed

The following classical two-point argument [17] states why source information alone may be insufficient. For two environments E_0, E_1 , let O include everything available to the decision rule, with laws P_0^O, P_1^O . Suppose acceptable risk-cost decisions form disjoint

sets D_0, D_1 , and loss satisfies $L_i(a) \geq \lambda \mathbf{1}\{a \notin D_i\}$ for $\lambda > 0$. Then every common decision rule satisfies

$$\max_i \mathbb{E}_i L_i(A) \geq \frac{\lambda}{2} (1 - \text{TV}(P_0^O, P_1^O)). \quad (4)$$

Indeed, deciding whether $A \in D_0$ defines a test of the environment; its sum of error probabilities is at least $1 - \text{TV}$. If a common expensive endpoint is acceptable, disjointness fails and the bound can be vacuous. For random queries, O must include the query jointly with its transcript. Our measurements establish response differences, not an estimate of full-observation total variation or an impossibility result for every policy. The bound explains the information question; the audit below supplies the operational test.

3.3 Certifying a Fixed-Target Decision

For a policy fixed independently of n i.i.d. target-certification queries, x failures yield the one-sided Clopper–Pearson bound [7]

$$U(x, n; \alpha) = \begin{cases} \text{Beta}^{-1}(1 - \alpha; x + 1, n - x), & x < n, \\ 1, & x = n. \end{cases} \quad (5)$$

Acceptance requires $U \leq \delta$. Confidence error α and allowed query risk δ are distinct. At zero failures, $U = 1 - \alpha^{1/n}$; $\alpha = \delta = 0.05$ requires at least 59 queries for acceptance to be possible. With 500 certification queries, nonzero failures can be accepted, but rejection can still reflect limited power near the boundary.

PROPOSITION 3.1 (JOINTLY QUALIFIED SELECTION ON ONE TARGET). *Freeze K policies before certification, and allocate $\alpha_j > 0$ with $\sum_{j=1}^K \alpha_j \leq \alpha$. On a common i.i.d. certification sample, compute $U_j = U(x_j, n; \alpha_j)$. Select any policy with $U_j \leq \delta$, or abstain if none qualifies. Then*

$$\Pr\{a \text{ policy is selected and } \mathcal{R}_G(\pi_{\text{selected}}) > \delta\} \leq \alpha. \quad (6)$$

PROOF. By a union bound, $\Pr\{\exists j : \mathcal{R}_G(\pi_j) > U_j\} \leq \sum_j \alpha_j$. Otherwise every selectable policy satisfies the risk limit. Independence among policies' bounds is unnecessary. $\square$

This is simultaneous risk control [2]. Selection may define the policies using a separate role; the proposition then applies conditionally on selection. A fallback must be included or separately justified. The statement is per fixed target, not simultaneous over a campaign, and does not bound risk conditional on the event of acceptance. The i.i.d. query interpretation concerns the represented workload; fixed-profile replay by itself proves no guarantee for a changed query population.

3.4 Why Pooling Is Not the Same Certificate

For m source builds and one target with exchangeable budget labels at a fixed query, conformal rank reasoning [1] gives a different guarantee. Let $r = \lceil (1-\eta)(m+1) \rceil$. Use the r th source order statistic if $r \leq m$ and it is finite; otherwise abstain. Then

$$\Pr\{a(q) < \infty, B_t(q) > a(q)\} \leq \frac{m+1-r}{m+1} \leq \eta. \quad (7)$$

This is build-marginal label coverage for a fixed query. It translates to execution failure for stable-tail labels, but not automatically for first-passing labels. Nine sources offer rank resolution 1/10, and old/refreshed builds need not be exchangeable. Accordingly, TCP's

nine-history replay derives its target qualification from Equation (6), not from a 5% conformal claim. The short supplement supplies the rank proof and a margin-based selection result, and states which assumptions are uninstantiated in the experiment.

4 ICBA and Recovery Routes

4.1 The Audit Contract

ICBA leaves graph construction and native traversal unchanged. Its input contract names the source and target snapshots, implementation, query population, action grid, permitted observations, failure event, and reference policy. Its output records raw reuse, the selected action, certification counts and bounds, actual fallback, execution risk, and lifecycle cost. These fields also expose when two experimental settings cannot be quantitatively combined.

Target information has three separate uses. Selection chooses a policy using its assigned queries. Certification checks that frozen policy and any fallback; it does not retrain or reorder them. Evaluation measures the completed decision, including fallback executions. A raw-reuse row is retained even if it would fail certification, so a successful endpoint cannot hide an unsuccessful transferred policy. If no policy qualifies, the audit outputs abstention; diagnostic endpoint execution is recorded as unqualified, not approved deployment.

4.2 TCP: History Pooling Followed by One Target Shift

The recovery experiment uses native HNSW actions

$$\mathcal{A} = (10, 20, 40, 80, 120, 160, 200). \quad (8)$$

For each query and old build, encode the first-passing grid index as $\tilde{j}_g(q)$; if none passes, store J while retaining the unresolved flag. For target seed t , pool the nine old builds other than the matched seed:

$$j_0^{(t)}(q) = \max_{g \in \mathcal{H}_t} \tilde{j}_g(q), \quad \pi_s(q) = a_{\min\{j_0^{(t)}(q)+s, J\}}. \quad (9)$$

The same-query old profiles exist for all three target roles. Thus TCP operates on a recurring, profiled workload, not unseen-query budget prediction. Storing a pooled index costs $O(Q)$ for Q profiles; a serving decision is a lookup, addition, and clipping before native search. Acquiring those profiles is a separate, potentially dominant expense.

Target selection scans $s = 0, \dots, J - 1$ in ascending order and takes the first shift whose selection-role CP bound at $\alpha = 0.05$ is at most 0.05. If none passes, it returns the endpoint shift. The scan fixes an ordered candidate before certification; it does not optimize NDC or certify the selected shift. Both the candidate and endpoint are then frozen.

Throughout this paper, TCP denotes the *target-calibrated pooling replay* in Equation (9). Its guarantee is determined by the target selection and certification procedure below, not by an exchangeable-build assumption.

4.3 Original Decisions and a Joint-Error Sensitivity

The original replay checks the selected candidate and endpoint separately at $\alpha = 0.05$ each. It accepts the candidate if its bound passes, otherwise executes the endpoint with the endpoint's qualification flag. The two allocations sum to 0.10; their individual 95% bounds therefore do not yield 95% joint control for the completed choice.

We add a *post-hoc frozen-response sensitivity*: keep every selected shift unchanged, use $\alpha_c = \alpha_e = 0.025$, and apply the same candidate-first order. If neither policy passes, abstain. This instantiates Proposition 3.1 at total $\alpha = 0.05$ under its query-sampling assumptions, separately for each target. Selection, the action class, and evaluation remain fixed; the analysis changes only the confidence allocation. We report it beside the original decisions as a sensitivity rather than a preregistered replication.

Figure 2 shows the data flow. The audit is reusable independently of TCP: a learned termination rule or a fixed budget can occupy the candidate input, provided its observation contract and failure event are retained.

4.4 A Source-Derived Slack Bridge

The second recovery policy removes TCP's target-selection step. Its source-design role of 375 queries per build is the previously analyzed S3 role; only the subsequent 500/500 roles are prospective. The supplement's fresh-query detail identifies the corresponding role manifest and freeze record. Before any of those 1,000 future-query vectors or truths were accessed, the frozen source role scanned the implementation's six native actions in ascending order. It sets a_s to the lowest action whose one-sided Clopper-Pearson bound is at most $\delta = 0.05$ under a Bonferroni allocation of $\alpha = 0.05/6$; if no action passes, it uses the endpoint. This rule fixes one action for each of the 24 sources in each implementation-dataset block. The registered +0, +1, +2 lanes then shift a_s upward by the stated number of native grid levels and clip at the endpoint.

On 500 previously untouched source-certification queries, each frozen lane and its endpoint receive error allocations $\alpha_c = \alpha_e = 0.025$. A source decision executes the lane only when both bounds are at most $\delta = 0.05$; otherwise it falls back to the certified endpoint. Another 500 untouched queries evaluate the completed source-to-target decision over all non-diagonal build pairs. The source actions, role identifiers, lane definitions, and their preregistration digest were fixed before any target-evaluation vector or truth was accessed.

This first bridge uses source information for design and certification; target truth appears only in the held-out portability audit. The three lanes are separate preregistered estimands. This stage measures whether a fixed slack transfers within a registered implementation-grid family, while reserving deployment qualification for the target-certified stage.

After that result and the manuscript evidence seal, the Faiss +1 lane was frozen for an independent target-certification replication. Per dataset, one shared 500-query role checks the candidate and endpoint on every target build with error allocations $\alpha_c = \alpha_e =$

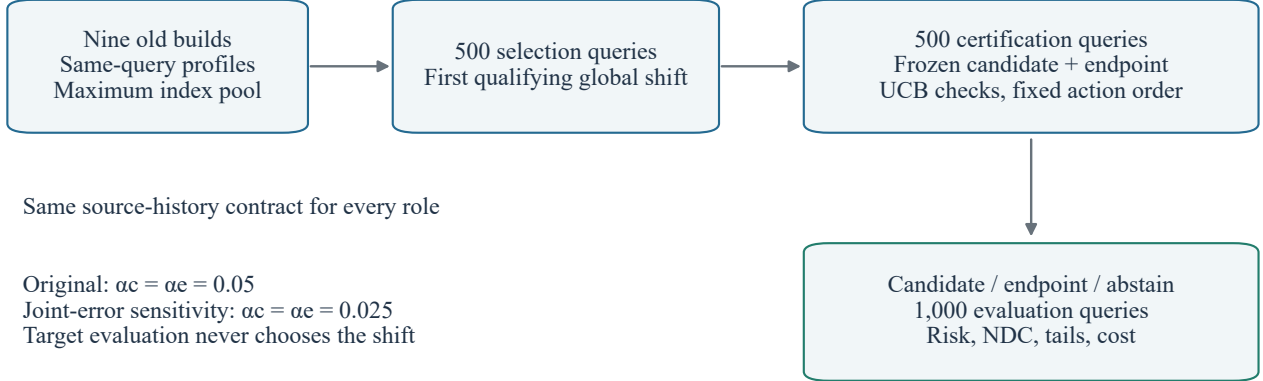


Figure 2: ICBA applied to TCP. Same-query old-build profiles define the pooled policy for each target role. Selection chooses one global shift; certification checks the frozen candidate and endpoint; evaluation has no feedback path. The original replay assigns 0.05 to each check. The joint-error sensitivity assigns 0.025 to each, allowing a per-target union bound of 0.05.

0.025; a disjoint 500-query role then evaluates the resulting decisions. Each role is reused across the 23 registered source histories of a target, producing 552 directed decisions without pretending that they are independent query samples. Candidate construction uses neither role, and evaluation cannot change the candidate, threshold, grid, or fallback. The decision abstains if the endpoint fails, deploys the candidate if its target UCB is at most 0.05, and otherwise executes the certified endpoint. This second stage tests a target-qualified fixed candidate, not TCP and not a newly target-selected action.

5 Experimental Design and Inference

Graph-only rebuilds. Each dataset contains 100,000 base vectors. SIFT-100K uses dimension 128 and Euclidean distance; Arxiv-Nomic-100K uses dimension 768 and normalized inner-product search. Each hnsplib dataset has 24 builds crossing eight seeds and three insertion orders (random, ascending local intrinsic dimensionality, and descending). The clean Faiss replay has 24 registered builds per dataset. Each uses 750 evaluation queries, all 552 non-diagonal directed build pairs, and unchanged base membership and truth. The hnsplib grid is (10, 20, 40, 80, 120, 200); the Faiss grid is (16, 32, 64, 128, 256, 512). Their native actions are not equated numerically.

For this study, the source response rule chooses $B_s^{\text{first}}(q)$, or the maximum action if unresolved. The target reference chooses $B_t^{\text{first}}(q)$ when finite. Its failure is the target-unresolved event; transport failure is either that event or failure of the actual target response at the source-derived action. This convention preserves unresolved mass rather than dropping difficult queries. It is a retrospective response experiment, not a deployable oracle algorithm or an independently source-certified policy.

Refresh and executable-policy replay. A separate seed-991 membership change replaces 5% of each 100K collection with previously excluded vectors, preserving cardinality. Truth is refreshed. Ten

old and ten refreshed builds per dataset are replayed through the official DARTH Faiss-HNSW fork, using graph degree parameter $M = 16$, construction $ef = 100$, one search thread, $k = 10$, and Equation (8). For each matched pair, TCP excludes that seed from its nine-history pool. The old-to-new diagnostic executes the *same* unshifted policy on both snapshots. Old-side certification checks the pre-existing pooled policy; no source outcome is used to select a different pool.

Target roles contain 500 selection, 500 certification, and 1,000 evaluation queries, with disjoint content and the same IDs reused across builds. Checks against base vectors and insertion reserves accompany the role manifest. Source-history responses for each role precede target selection. Reusing historical profiles for evaluation IDs is part of the recurring-workload contract, not a learned predictor’s held-out generalization test. Graph construction, membership refresh, and statistical resampling use separately recorded seed roles.

Fresh-query source-only boundary. After the preceding analyses were frozen, 1,000 previously untouched query IDs per dataset were split in registered order into 500 source-certification and 500 target-evaluation queries, with zero overlap with the earlier 375-query source-design role or any prior evaluation role. Only these 500/500 roles are prospective. We reuse the same 24 hnsplib and 24 Faiss indexes per dataset; no graph is rebuilt or mutated. The source rule, its 96 actions, the +0, +1, +2 lanes, and the role identifiers were sealed before accessing future-query vectors or truth. Candidate and endpoint receive $\alpha = 0.025$ each on the source; target evaluation cannot change the action or fallback. Query-bootstrap intervals retain each query’s complete 552-direction vector. This stage audits source-only portability conditional on the registered builds.

Post-seal target-certified replication. The positive Faiss +1 candidate was then fixed without retuning. For each dataset, two additional roles of 500 queries were drawn from a previously unused source-data window and frozen before vector or truth access. They have zero overlap with one another and all earlier registered query roles. The same certification role is evaluated on each target build and reused across its 23 source histories; the disjoint evaluation role is handled analogously. Target certification applies $\alpha_c = \alpha_e = 0.025$ to the frozen candidate and endpoint. Primary 95% intervals use a crossed bootstrap that independently resamples the 24 target builds and 500 shared evaluation-query IDs 5,000 times with seed 991 while retaining all source directions inside each target–query cell. Target-only and query-only intervals are sensitivity analyses. Fixed-machine interleaved timing is reported only as an exploratory sensitivity; NDC remains the primary endpoint.

Deep1M scale confirmation. The scale replication reuses eight registered hnswnlib Deep1M indexes crossing four seeds with two insertion orders. Its source-derived one-rung candidate is frozen before three disjoint 500-query roles are accessed: source certification, target certification, and evaluation. The target stage applies the same $0.025 + 0.025$ candidate/endpoint allocation and keeps all 56 directed source–target decisions grouped within their eight targets. The crossed bootstrap resamples targets and shared query IDs; LOTO deletes a target without refitting. No parameter, lane, or endpoint is changed after the 100K results.

Metrics. Risk is query failure under $k = 10$, $\tau = 0.95$. Relative serving gain is

$$g = 1 - \frac{\bar{c}_{\text{policy}}}{\bar{c}_{\text{endpoint}}}, \quad (10)$$

a ratio of pooled means, equivalently a relative total-NDC reduction for equal query counts. We report query-pooled p95/p99 NDC and per-target tails, retaining every fallback execution. These are search-work measures, not latency measurements. Observed tail non-increase is distinguished from a formal noninferiority test.

Inference and its sampling units. Graph-only intervals resample 750 query IDs, retaining each query’s complete directed-pair vector, with builds fixed. They do not treat 552 overlapping directions as independent builds. For the refresh replay we recompute 5,000 crossed bootstrap replicates with seed 991: sample target-build indices, then sample one shared query-index vector used in *all* sampled builds. The paired policy/reference outcomes remain together. Percentile intervals reflect this product resampling, conditional on the stored source histories and selected shifts; they are not confidence guarantees for a new build-generation population. Build-only and shared-query-only sensitivities appear in the supplement.

LOTO deletes a target without refitting, and deletion of the largest relative-gain target is one specified member of that sensitivity family. Acquisition-cost intervals resample target builds conditional on the finite profiling/query records. Vamana and external-method intervals retain their original units, identified beside the corresponding estimates. No interval is pooled across these protocols.

Evidence order. We answer nine questions in reading order: graph-only transfer (RQ1), matched refresh transfer (RQ2), audit decisions (RQ3), recurring-profile and fixed-slack recovery (RQ4), tails (RQ5), build sensitivity (RQ6), external and graph-family scope (RQ7), economics (RQ8), and reproducibility (RQ9). The source-only boundary, Faiss target-certified replication, and Deep1M scale confirmation use separately frozen, disjoint roles. Derived analyses retain the original responses and decisions.

6 Where Budget Portability Fails

6.1 RQ1: Do Responses Transfer When Only the Graph Changes?

Retrospective first-passing actions frequently fail after a graph-only rebuild, even though collection membership, queries, and truth are unchanged. Table 1 reports absolute target risk, the unresolved-reference risk, and their difference. Across hnswnlib and Faiss on both datasets, the increment ranges from 17.17 to 23.60 percentage points; every query-cluster interval is above zero. This is a response-level result independent of TCP’s pooling or recalibration.

The variation is principally among finite budgets: 75.87–91.07% of queries have different finite first-passing actions across builds, while only 0.53–2.67% switch between resolved and unresolved states. Removing the 1% of queries with largest incremental contribution leaves increments of 21.37/16.95 percentage points for hnswnlib and 23.43/17.53 for Faiss, on SIFT/Arxiv respectively. Endpoint censoring and a handful of extreme queries therefore do not explain the result.

This experiment diagnoses the budget–response relation using retrospective truth on both source and target. It establishes that source success alone is insufficient evidence for action reuse. The measured mechanism is behavioral variation across builds; structural attribution to particular edges or navigation paths lies outside this response-level design.

6.2 RQ2: Does an Executable Policy Deteriorate after Refresh?

The matched old/new replay holds the nine-history policy fixed and applies it to both snapshots. Unlike RQ1, this changes 5% of the vectors as well as rebuilding. SIFT’s empirical risk rises from 3.77% to 6.69%; Arxiv-Nomic’s rises from 3.19% to 7.24%. The corresponding paired increments [crossed 95% interval] are 2.92 [1.75, 4.07] and 4.05 [2.86, 5.31] percentage points. Thus this is a measured deterioration, not merely an absolute target risk with an unknown source baseline.

Independent single-policy CP checks at $\alpha = 0.05$ qualify six of ten old SIFT policies and five of ten old Arxiv policies. All eleven fail the corresponding target check. This paired diagnostic uses per-policy checks rather than simultaneous control over the eleven transitions. Target evaluation risk exceeds 5% in all ten builds of each dataset; removing the highest-risk target leaves 6.61% and 7.13%. The deterioration is distributed across targets.

The same policy information is held fixed in each old/new comparison, but the experiment does not separate changed graph paths from changed nearest-neighbor membership. Together, RQ1 and

Table 1: Graph-only response transfer at $k = 10$, $\tau = 0.95$. Each row has 24 builds, 552 directed pairs, and 750 shared query IDs. Risk entries and increments are percentages or percentage points [95% query-cluster interval], conditional on the fixed build set. The reference uses the target’s own first-passing response; its failures are unresolved queries. Finite variation counts queries with at least two distinct finite first-passing actions, divided by all 750 queries.

Implementation	Dataset	Absolute risk [95% CI]	Reference risk [95% CI]	Increment [95% CI]	Finite variation
hnsplib HNSW	SIFT	22.31 [21.51, 23.13]	0.77 [0.41, 1.19]	21.55 [20.76, 22.33]	89.33
Faiss HNSW	SIFT	23.67 [22.86, 24.46]	0.07 [0.01, 0.19]	23.60 [22.78, 24.39]	91.07
hnsplib HNSW	Arxiv	18.24 [17.31, 19.20]	1.07 [0.54, 1.66]	17.17 [16.33, 18.06]	75.87
Faiss HNSW	Arxiv	17.93 [17.03, 18.85]	0.17 [0.07, 0.29]	17.77 [16.86, 18.68]	75.87

RQ2 establish complementary facts: unchanged data do not make the response function portable, and a recurring-query policy can lose its old qualification after refresh plus rebuilding.

7 Auditing and Recovering Rebuilt Workloads

7.1 RQ3: What Does the Audit Actually Authorize?

Auditing unshifted source reuse rejects every candidate and routes all 20 target decisions to the endpoint in the original replay. The audit therefore prevents cheap but unqualified reuse, yet recovers no search saving by itself. Target-selection recalibration changes the candidate before certification. The original procedure accepts 19 of 20 recalibrated candidates; Arxiv seed 1229 has candidate UCB 0.05056 and falls back despite being close to the threshold.

The joint-error sensitivity is more selective (Figure 3). With $\alpha_c = \alpha_e = 0.025$, it accepts all ten SIFT candidates and six Arxiv candidates. All twenty endpoints pass their allocated checks, so four Arxiv decisions fall back and none abstains. The completed choice satisfies the per-target construction in Proposition 3.1 under its query-sampling assumptions. A rejection routes execution to the endpoint because the 500-query certificate does not qualify the candidate.

7.2 RQ4: How Much Search Efficiency Is Recovered?

Recalibration restores substantial search efficiency within the recurring-profile workload (Table 2). The original decisions reduce risk to 1.79% and 2.04%, with relative mean-NDC reductions of 44.39% and 44.79%. Crossed resampling widens SIFT’s interval compared with the old independently-within-build procedure: it is now [42.76, 46.01]%, rather than [43.88, 44.91]%. Preserving shared query dependence changes uncertainty even when the point estimate is unchanged.

Under the joint-error allocation, SIFT is unchanged, while Arxiv risk becomes 1.68% and relative mean-NDC reduction becomes 29.86 [14.75, 45.00]%. Figure 4 shows that stricter qualification moves the decision toward the endpoint: it reduces risk and sacrifices some efficiency. Both datasets retain positive search-work reductions relative to the same graph’s fixed endpoint. Adaptive methods with different information budgets are examined as qualification studies in Section 8.

7.3 RQ4 Boundary and Replication: Can Fixed Slack Transfer?

The first fresh-query stage tests the source-derived policy of Section 4.4. Source certification determines the executable candidate; 500 held-out target queries provide a portability audit. The complete three-lane table remains in the supplement. On hnsplib, the +1 lane clips to the endpoint for all registered 100K sources and saves no work. On Faiss, the same lane retains 8.27% and 29.09% mean-NDC savings, with held-out empirical risks of 0.18% and 0.67%; no held-out direction exceeds 5% empirical risk. This source-only stage identifies a viable Faiss candidate family for independent target qualification.

The post-seal replication closes that gap with new, mutually exclusive target certification and evaluation roles (Table 3). Every candidate and endpoint has a target CP UCB below 5%; all 552 directions per dataset deploy the frozen candidate, with no fallback or abstention. The largest candidate certificate counts are 5/500 on SIFT and 6/500 on Arxiv-Nomic (UCBs 2.32% and 2.59%). Evaluation risk is 0.126 [0.040, 0.250]% on SIFT and 0.530 [0.233, 0.915]% on Arxiv-Nomic. Mean-NDC savings remain 8.283 [7.921, 8.557]% and 29.083 [28.653, 29.518]%. These crossed target–query intervals are wider than the target-only intervals but leave the conclusions unchanged. Pooled and worst-target p95 ratios remain below one, and leaving out any target preserves positive mean savings.

The candidate is below the endpoint in 92/552 SIFT directions and 322/552 Arxiv directions; in the others the upward rung clips to the endpoint. Thus every decision can deploy the qualified candidate while many isolated directions have zero saving. Fixed-machine interleaved timing points in the same direction (9.27% and 26.57% mean reductions); NDC remains the primary efficiency endpoint because truth and control time are absent from the timing ledger.

Scale confirmation on hnsplib. The preregistered Deep1M replication applies the same fixed-slack logic without retuning. All eight source candidates remain below the endpoint, and all 56 target decisions qualify the candidate with no fallback or abstention. Held-out risk is 1.632 [0.804, 2.614]%, mean-NDC saving relative to efSearch=1,200 is 34.282 [33.533, 35.045]%, and pooled p95/p99 ratios are 0.682/0.695. Every target remains positive, and the minimum LOTO saving is 34.126%. Together with the Faiss results, this establishes target-certified fixed-slack recovery across two HNSW implementations and 100K-to-1M scale. The 100K hnsplib endpoint-clipping result remains the counterexample: the route

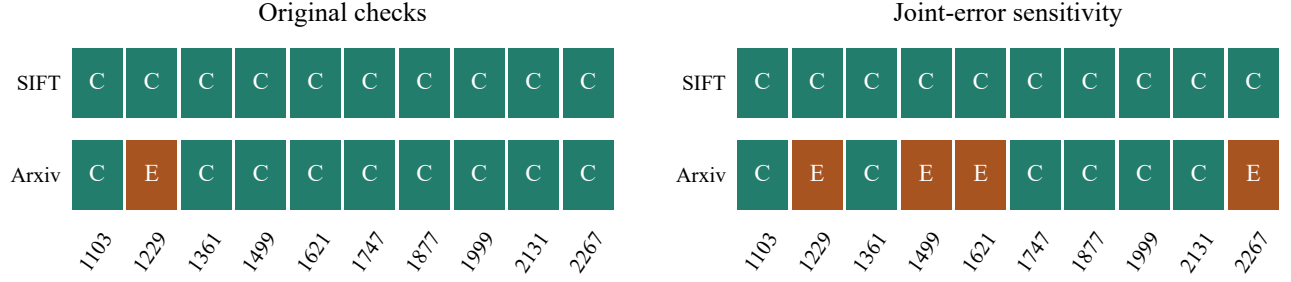


Figure 3: One cell per target decision, ordered by registered seed rather than a time axis. C: candidate accepted; E: endpoint fallback. Left: original two individual 95% checks. Right: post-hoc 0.025+0.025 joint-error allocation. The same selected shifts and certification queries are used in both. Candidate acceptance falls from 19/20 to 16/20, while every completed decision remains qualified by its corresponding endpoint or candidate check.

Table 2: Completed refresh decisions on ten targets and 1,000 evaluation queries per target. Query risk and relative mean-NDC reduction are percentages [crossed 95% interval]; intervals are conditional on fixed source histories and selected shifts. Original and joint-error rows differ only in certification composition. All fallback executions are included. Mean and p95/p99 are distance calculations, not latency.

Dataset	Decision	Risk [95% interval]	Reduction [95% interval]	Mean NDC	p95	p99
SIFT	Endpoint	0.97 [0.53, 1.50]	0.00 [0.00, 0.00]	1,826	2,615	2,855
SIFT	Direct reuse	6.69 [5.65, 7.77]	59.51 [57.83, 61.20]	739	2,132	2,706
SIFT	Original recalibration	1.79 [1.25, 2.40]	44.39 [42.76, 46.01]	1,015	2,349	2,774
SIFT	Joint-error replay	1.79 [1.25, 2.40]	44.39 [42.76, 46.01]	1,015	2,349	2,774
Arxiv	Endpoint	1.01 [0.61, 1.47]	0.00 [0.00, 0.00]	2,397	3,252	3,504
Arxiv	Direct reuse	7.24 [6.18, 8.36]	64.21 [62.51, 65.85]	858	2,586	3,322
Arxiv	Original recalibration	2.04 [1.45, 2.69]	44.79 [34.32, 50.96]	1,323	2,974	3,398
Arxiv	Joint-error replay	1.68 [1.10, 2.35]	29.86 [14.75, 45.00]	1,681	3,117	3,460

Table 3: Post-seal Faiss target-certified replication. Each dataset has 24 targets, 552 directed decisions, 500 target-certification queries, and 500 disjoint evaluation queries. C/F/A counts candidate, fallback, and abstention decisions. Risk and mean-NDC saving are percentages [95% crossed target-build/query-ID bootstrap interval]. P95 reports pooled/worst-target ratios to the endpoint; LOTO is the minimum mean saving after deleting one target.

Dataset	C/F/A	Eval. risk [95% CI]	NDC saving [95% CI]	P95 ratio	LOTO saving
SIFT	552/0/0	0.126 [0.040, 0.250]	8.283 [7.921, 8.557]	0.9964 / 0.9987	8.267
Arxiv	552/0/0	0.530 [0.233, 0.915]	29.083 [28.653, 29.518]	0.9797 / 0.9828	29.027

is useful when the registered source candidate leaves recoverable slack.

7.4 RQ5: Does Recovery Worsen the Search-Work Tail?

Neither recalibration rule increases query-pooled p95 or p99 relative to the endpoint. For joint-error replay, p95 is 2,349 versus 2,615 NDC on SIFT, and 3,117 versus 3,252 on Arxiv; p99 is 2,774 versus 2,855 and 3,460 versus 3,504. Every individual target also has non-increasing p95 and p99. Figure 5 displays the per-target ratios without connecting unrelated seeds.

The fallback is a build-level choice, so its serving distribution is the fixed endpoint’s distribution for that build. Combining fallback and candidate-serving builds can change pooled quantiles through composition alone. We therefore report pooled and per-target tails separately from the larger mean gain.

7.5 RQ6: Is the Benefit Driven by One Target?

The minimum LOTO mean-NDC reduction is 44.37%/44.23% for original SIFT/Arxiv decisions, and 44.37%/27.64% for joint-error decisions. Deleting the largest relative-gain build produces those

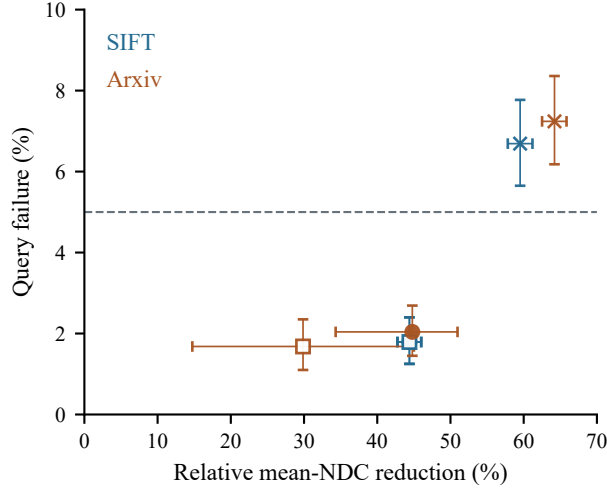


Figure 4: Refresh risk–work tradeoff with crossed 95% intervals. The 5% line is the query-risk target, not a confidence bound. Filled symbols: original recalibration; open symbols: joint-error replay; crosses: direct reuse. SIFT’s original and joint points coincide.

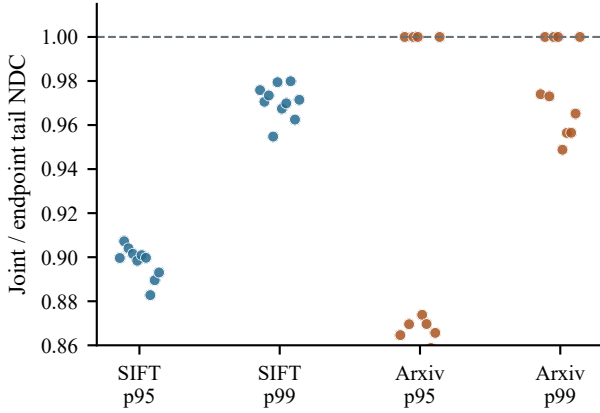


Figure 5: Per-target tail ratios for joint-error replay, relative to that target’s endpoint; ten dots per dataset/quantile. Equality reflects an endpoint fallback. All ratios are at most one. The figure reports observed NDC tails rather than a formal tail certificate.

same minima, so the two checks overlap. Target-only and shared-query-only interval sensitivities in the supplement separate the uncertainty sources: SIFT varies mainly across queries, whereas Arxiv’s stricter result also varies with candidate/fallback composition across targets. Every registered deletion preserves positive recovery on the fixed target set.

8 RQ7: Scope across Methods, Scale, and Graph Family

External adaptive methods. The frozen DARTH adapters produce target query-failure rates of 22.55% on SIFT-100K and 23.12% on Arxiv-Nomic-100K at Recall@10 < 0.95. Ada-ef yields 9.52 [9.44, 9.60]% on ten Arxiv targets, with a target-build bootstrap conditional on its query set. All ten Ada-ef candidate certificates fail; audited execution uses the qualified endpoint and has zero search gain relative to that endpoint. Ada-ef is absent from Euclidean SIFT because the inspected estimator implements cosine/inner-product semantics, not a semantics-preserving L2 path. Its exclusion is an implementation-scope boundary, not a failed SIFT result.

These measurements show why query-risk qualification matters beyond TCP. Without matched source checks, they do not isolate rebuild-induced loss; their failure event also differs from mean recall and expected false-negative fraction. We therefore use them as qualification studies with their own information budgets.

Larger scale. On Deep1M, eight hnsplib builds cross four seeds with two insertion orders, giving 56 directed pairs on 1,000 queries. Source-derived first-passing actions have 28.86% transport risk against 8.82% target-reference risk. The incremental risk is 20.04 [17.29, 22.77] percentage points, using a target-build bootstrap conditional on the frozen source histories and query set. This is additional loss above a difficult, partly unresolved target reference. Section 7.3 reports the separate 500/500/500-query target-certified recovery stage on these indexes.

Vamana-style evidence. Six source and six separate target builds produce 36 directions and 750 queries per dataset, using native search-list size and beam width one. The frozen event table combines its stage-specific under-budget and censoring outcomes. A target-build reanalysis gives 16.86 [15.88, 17.93]% on SIFT and 11.37 [10.80, 11.99]% on Arxiv. Its recorded responses do not reconstruct exactly the first-passing execution estimand used for hnsplib/Faiss. We retain it as directional cross-family evidence, not another row of Table 1.

Vamana cost-tax intervals include zero: 1.07 [−1.03, 2.86]% and 1.17 [−0.83, 3.03]%, respectively, for the mean of per-target ratios in that reanalysis. Thus it supplies no established positive cost penalty or cross-engine superiority. Table 4 summarizes which question each setting can answer.

Together with RQ1, these results broaden the evidence without collapsing unlike estimands. Response non-portability appears across HNSW implementations and at larger scale; qualification changes the executable outcome of adaptive actions; and Vamana supplies directional graph-family evidence under a narrower event definition. The common conclusion is the need to bind a budget decision to its target index and risk contract.

Table 4: Scope of the extension evidence. Each row retains its event, reference, and inference unit. All numeric intervals in the text are 95%; no common cross-engine ranking or pooled effect is formed.

Setting	Event and reference	Supported comparison	Unit / query count
DARTH	Absolute target Recall @10 < 0.95 risk; no matched reference subtraction	Target qualification under the frozen adapter	Target summaries; 1,000/build
Ada-ef	Absolute target Recall @10 < 0.95 risk; endpoint for audited saving	Candidate rejection versus endpoint execution on Arxiv	10 target builds; 1,000/build
Deep1M	Transport versus target reference; certified slack versus efSearch 1200	Transfer loss and certified recovery at 1M scale	8 targets; 1,000 prior + 1,500 fresh
Vamana-style	Frozen stage under-budget/censoring event; stage-specific cost reference	Directional cross-family transfer evidence, not a harmonized HNSW increment	6 target builds; 750 shared queries

9 RQ8: When Does Recovery Repay Its Information Cost?

Search savings repay recovery only over a sufficiently long recurring workload. We express the accounting in native distance evaluations,

$$C_{\pi}(N) = C_{\text{acquire},\pi} + N\tilde{c}_{\pi}, \quad N_{\text{BE}} = \frac{\Delta C_{\text{acquire}}}{\tilde{c}_{\text{end}} - \tilde{c}_{\pi}}. \quad (11)$$

If incremental acquisition is positive but serving saving is not, no finite break-even exists. Graph rebuilding is common to both policies on the same target. Control adds no distance calls but is not assigned zero elapsed time.

Two explicit acquisition scenarios. With *cached history*, old query profiles and their truth are sunk costs. We charge the complete target-selection grid, the selected candidate’s certification searches, additional endpoint-certification searches where the actions differ, and exact target truth for the 1,000 selection/certification queries. Truth is modeled as one scan of 100,000 vectors per query. With *cold complete history*, we also charge sequential old-build profiling for all three roles on all nine contributing histories and exact old-snapshot truth for all 2,000 query IDs. That truth is reusable across old builds of the same collection and is counted once, not nine times. These are workload-acquisition scenarios, not newly measured elapsed-time costs.

We conservatively charge all endpoint certification to recovery, although the reference may also need some of this work. We price the evaluated decision rule without charging repeated calls for the same action and query twice. The earlier partial calculation included old profiling only for evaluation IDs and omitted old truth and extra endpoint certification. Completing these items materially changes the economic conclusion.

For joint-error replay, cold-history break-even is approximately 396,000 executions on SIFT and 457,000 on Arxiv, versus 129,000 and 148,000 with cached history (Table 5). Four Arxiv targets fall back and never repay a positive recovery overhead through serving savings. The original procedure has lower cold-history break-even on Arxiv, approximately 305,000, but different certification composition. It is therefore insufficient to attach the original economics to the stricter decision rule.

At $N = 10^6$, the cold-history joint-error model yields mean net savings of about 490 million NDC on SIFT and 389 million on Arxiv.

Table 5: Joint-error recovery: break-even query executions in thousands [95% target-build interval], conditional on the finite acquisition workload. The ratio of mean overhead to mean saving is not a per-build guarantee. “No saving” counts nonamortizing targets.

Dataset	History	Break-even, 10^3	No saving
SIFT	Cached	129 [129, 129]	0/10
SIFT	Cold	396 [395, 397]	0/10
Arxiv	Cached	148 [99, 296]	4/10
Arxiv	Cold	457 [305, 915]	4/10

Their target-build intervals are [488, 491] million and [31, 747] million, respectively. At 10^5 , both cold-history means are negative. Figure 6 shows how the crossover depends on what information already exists.

Target-certified fresh-query policy. The post-seal replication has a different ledger. Per dataset, the same 500 certification queries are run on every target and reused across its 23 source histories. For each target, exact truth costs $500 \times 100,000$ distances; measured candidate and endpoint searches are added with identical actions deduplicated. Pairwise accounting charges this target audit to every directed source–target unit, whereas shared-target accounting charges it once and reuses it over the 23 registered sources (Table 6). For example, SIFT pairwise acquisition sums to 30.588 billion NDC and serving saving to 457,122.72 NDC per portfolio execution, giving $30.588\text{B}/457,122.72 = 66,914$ executions. Shared-target acquisition is 1.380 billion NDC over the same serving saving, giving 3,019 executions. The corresponding Arxiv thresholds are 17,018 and 755. Because clipping gives zero saving in 460/552 and 230/552 directions, the pairwise ratios are portfolio thresholds, not per-direction guarantees.

At $N = 10^6$ per direction, mean shared-target net saving per target (aggregated over its 23 source directions) is 18.99 billion NDC on SIFT and 77.61 billion on Arxiv; both target-build interval lower bounds, LOTO minima, and delete-largest-target checks remain positive. This closes the *target-stage* NDC ledger. It does not close complete cold lifecycle economics: historical source-policy acquisition has no compatible per-query NDC record and is not assigned zero. Exact-truth/control wall time is also absent, so search-only timing break-even remains diagnostic rather than an end-to-end latency result.

Table 6: Target-stage NDC ledger for the target-certified Faiss candidate. Break-even is a ratio of summed acquisition cost and serving saving [95% target-build bootstrap interval]. “No saving” counts units with zero serving reduction; the final column is the first registered horizon with positive aggregate net NDC.

Dataset	Accounting	Units	Break-even [95% CI]	No saving	First positive N
SIFT	Pairwise	552	66,914 [64,805, 69,940]	460	100,000
SIFT	Shared target	24	3,019 [2,923, 3,157]	0	10,000
Arxiv	Pairwise	552	17,018 [16,767, 17,276]	230	100,000
Arxiv	Shared target	24	755 [744, 767]	0	1,000



Figure 6: Joint-error recovery, modeled net distance savings per target versus serving horizon. Solid: cold complete history; dashed: cached history. Dots mark reported horizons; lines depict the stated affine cost model, not additional measurements. Endpoint-fallback targets remain in the average.

This calculation assumes repeated executions over the profiled workload. A cache miss requires new information or a declared endpoint action; an unbounded stream of new query identities can invalidate the amortization. Acquisition of historical graphs themselves is not reconstructed here: cold history means acquiring profiles on retained graphs, not rebuilding the entire archive. Memory traffic, storage, energy, and control time require separate prices. The result is a conditional distance-work benefit, not an end-to-end latency or monetary claim.

10 Reproducibility and Operational Implications

10.1 RQ9: Can the Decisions and Numbers Be Checked?

The compact evidence package contains per-target certification failures, confidence allocations, selected shifts, paired evaluation arrays, cost components, fresh-query summaries, and generated table inputs. One script regenerates figures and typeset

rows; another reanalyzes frozen native-response files through explicit repository and replay-root arguments. Historical records, the source-only boundary, the Faiss target-certified stage, and Deep1M confirmation retain separate evidence labels and role manifests.

The artifact audit preserves 80 pre-audit scientific files, passes a neutral-path smoke and 28 focused tests, and exposes guarded replay entry points for DARTH, Vamana, and Deep1M. In a clean Python environment built from the anonymous allowlist alone, the manuscript package passes 244 earlier evidence checks, reproduces generated table text, regenerates six figures, and verifies 210/210 source-only fresh-query checks. The post-seal checker validates 1,104 Faiss target decisions, 48 target summaries, four cost summaries, and both cost ledgers. The Deep1M checker covers all 56 target-certified decisions and their crossed-bootstrap, tail, and deletion summaries. A fresh Faiss environment also reproduces 6,000 native action rows for one registered SIFT build with zero top- k , recall, NDC, or action-order mismatch. Full native replay requires the separately provisioned vectors, truth, indexes, binaries, and role manifests listed in the supplement.

Operational decision. The evidence yields a three-step maintenance rule. First, treat an inherited budget as source evidence and measure its target response. Second, qualify the completed candidate/fallback procedure on target queries under a declared risk event. Third, compare its serving savings with the cost of the information used to construct and certify it. ICBA records these steps. TCP is appropriate when recurring-query histories exist and a target selection role can choose a shift. Fixed slack uses a lower-information candidate and target certification; its Faiss and Deep1M results show that this route can retain work, while endpoint clipping on the 100K hnslib grids identifies a regime with no recoverable slack.

Scope of the claim. The portability estimates are conditional on the registered builds, query populations, action grids, and failure event. TCP’s positive results concern recurring profiles on two Faiss-HNSW refresh workloads. Fixed-slack target qualification succeeds on two Faiss 100K blocks and one hnslib Deep1M block; Vamana contributes directional transfer evidence under a different recorded event. The TCP ledger models cached and cold profile acquisition in NDC. The fixed-slack ledger covers target-stage acquisition, while historical source acquisition and end-to-end truth/control time remain unmeasured. The experiments establish response variation and qualified recovery, rather than edge-level causality, open-build-population coverage, or hardware-general latency gains.

11 Conclusion

Search-budget portability is an index-conditioned decision. Across graph-only rebuilds, source first-passing actions add substantial failure above target-specific references; after refresh, an unchanged recurring-query policy loses its old qualification. ICBA converts these observations into an auditable contract for candidate construction, target certification, fallback, evaluation, and cost. TCP recovers search work for recurring profiles, while fixed slack demonstrates target-certified recovery on Faiss and hnswlib Deep1M and exposes endpoint-clipped cases with no gain. The combined evidence shows how to audit reuse, recover work when target evidence supports it, and decide whether that recovery is worth acquiring.

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